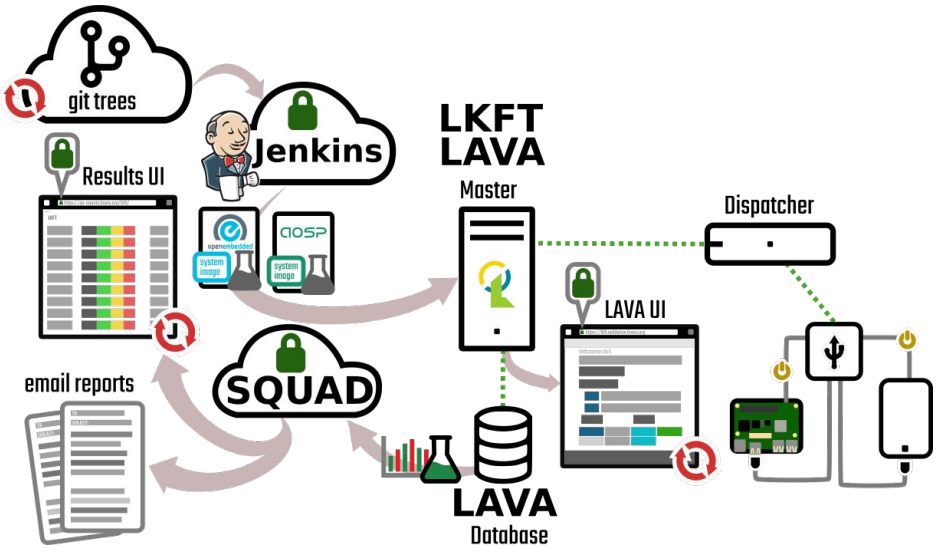
Linux Kernel Functional Testing (LKFT)

Functional Testing of Mainline, LTS and Android Common Kernels

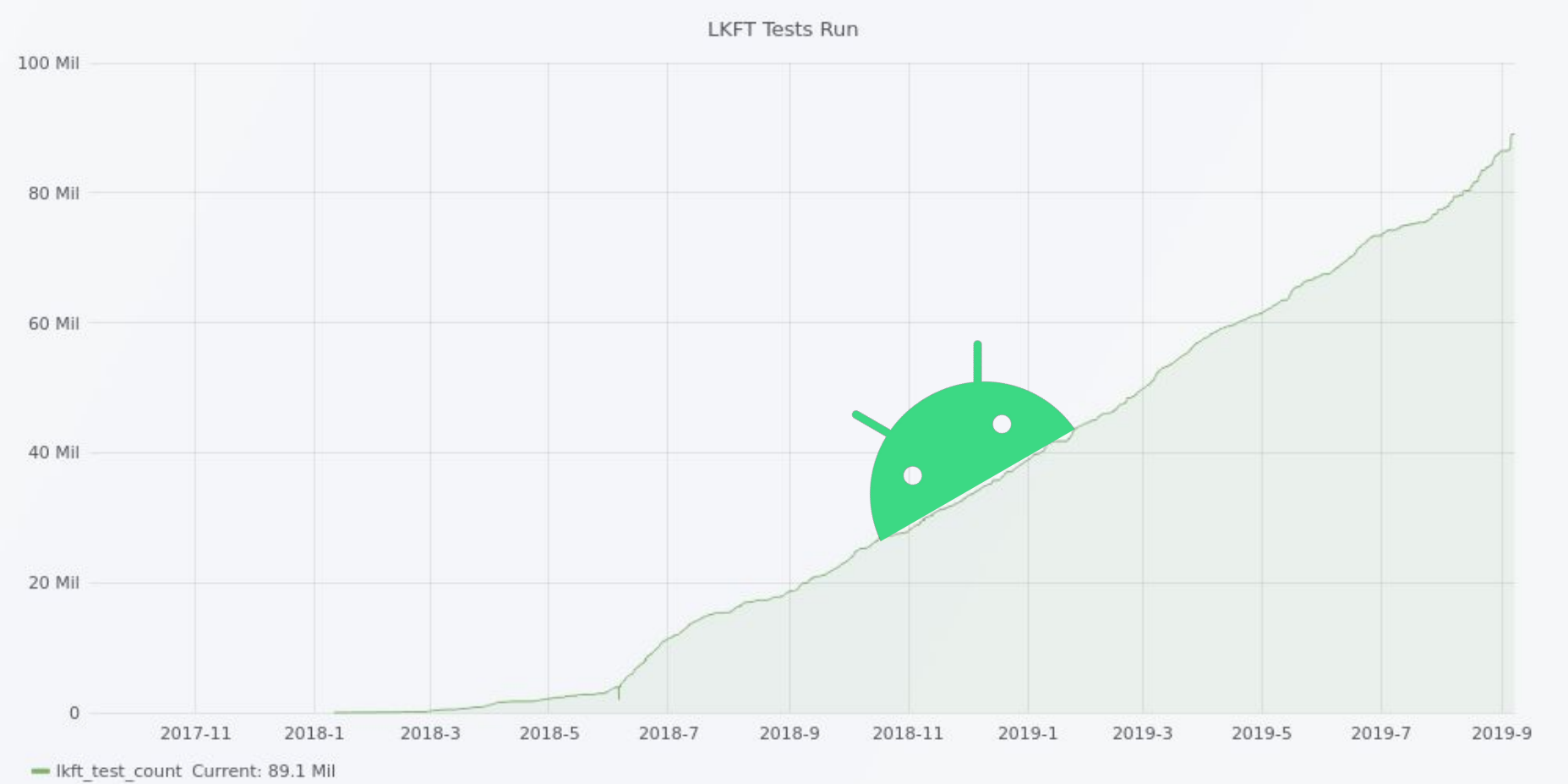
Linux Kernel Functional Test (LKFT) Components

The LKFT framework is a collection of software tools and hardware devices. LKFT is composed of

- Lab - test targets
  - You can host your own remote lab
- Jenkins - Builds -> on demand builders
- Linaro Automated Validation Architecture (LAVA) - Manage the DUTs
- SQUAD - Scheduling, data analysis and reporting framework
- Test definitions



Backed by Linaro's Kernel Validation team to maintain, triage results, and escalate



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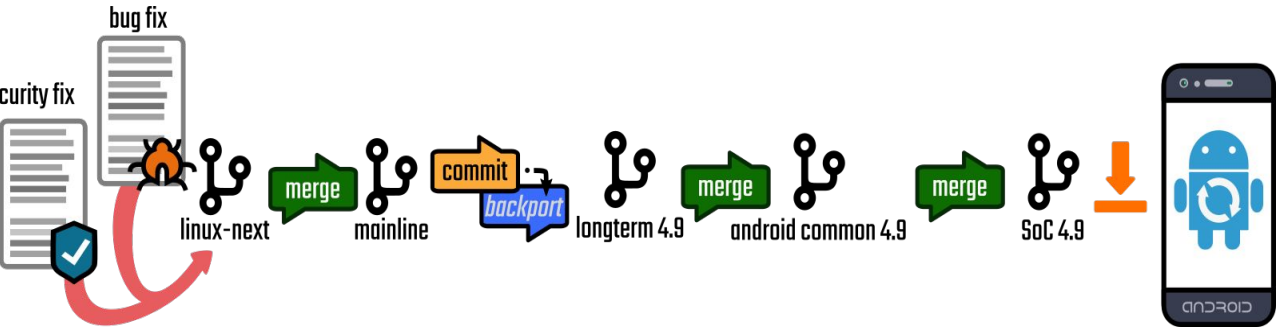
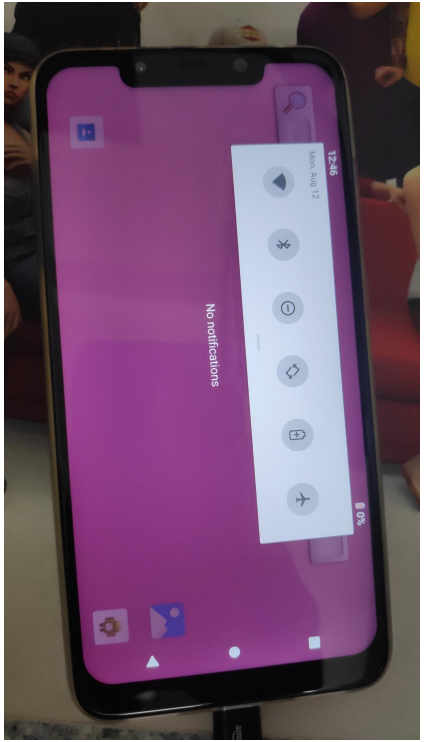
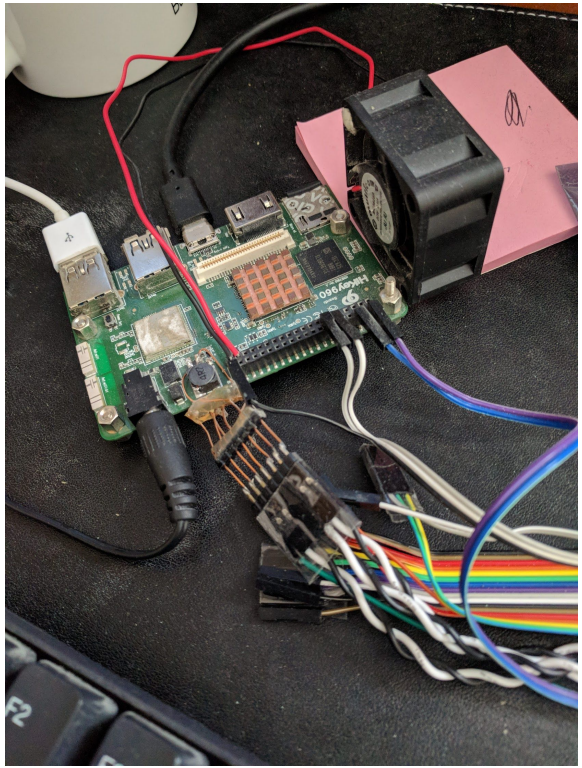
Mainline -> Stable -> Android Common Kernel



To the right is a Poco F1, booting Android with a mainline linux kernel (and a couple of patches but not many!) Pixel 3 boots mainline too! (with patches)

Being able to boot/run Android on mainline means we can test mainline, future stable and future LTS kernels at RC.



Boot testing is tired, Functional testing is wired

- CTS / VTS (kselftest, LTP)
  - Remember exercising the kernel is what we want
  - CTS pushes on Camera, BT, Networking
- Add energy probes, run benchmarks, workloads and look for kernel regressions in EAS
- Add a Chamelium board to capture video / sound
  - IGT
  - Synthmark
  - V4l2
  - libcamera
- Regressions get to be more than pass/fail, EAS regression uses statistical tests that spot anomalies